

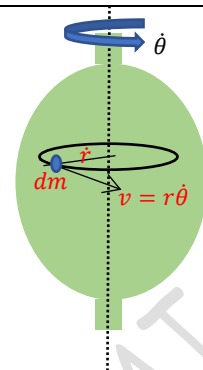
7.5 Work and Energy in Rotation

When a body rotates about a fixed axis with angular velocity $\dot{\theta}$, any particle of mass dm at a distance r will move along a circle of radius r and its velocity along the circular path $v = r\dot{\theta}$.

The kinetic energy of the particle $= \frac{1}{2} dm v^2 = \frac{1}{2} dm (r\dot{\theta})^2$

Total kinetic energy of the body, $T = \int \frac{1}{2} dm (r\dot{\theta})^2$

$$T = \frac{1}{2} \dot{\theta}^2 \int r^2 dm = \frac{1}{2} I \dot{\theta}^2$$



The equation of rotation of rigid body is $I\ddot{\theta} = \Sigma M$

$$I \dot{\theta} \frac{d\dot{\theta}}{d\theta} = \Sigma M \gg \gg I \dot{\theta} d\dot{\theta} = \Sigma M d\theta$$

Integrating both sides

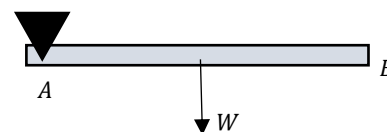
$$\int_{\dot{\theta}_0}^{\dot{\theta}} I \dot{\theta} d\dot{\theta} = \int_{\theta_0}^{\theta} \Sigma M d\theta$$

$$\frac{1}{2} I \dot{\theta}^2 - \frac{1}{2} I \dot{\theta}_0^2 = \int_{\theta_0}^{\theta} \Sigma M d\theta$$

This shows that the change in kinetic energy of a body rotating through an angle θ is equal to the work done by the external moment over the angular displacement θ

Example 1

If the slender prismatic bar AB is released from rest in the horizontal position as shown and rotates about A under the influence of gravity, find the reaction at A as it swings through the vertical position



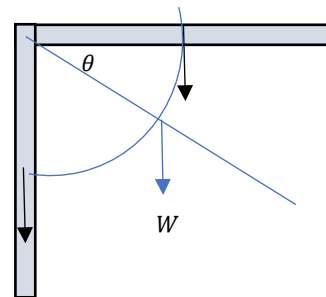
The bar starts from rest.

The initial kinetic energy of the bar $= 0$

Let $\dot{\theta}$ be the velocity of the bar when it swings vertically.

The final kinetic energy of the bar $= \frac{1}{2} I_A \dot{\theta}^2$

Change in kinetic energy $= \frac{1}{2} I_A \dot{\theta}^2 - 0$



Moment of the weight about the axis of rotation $M = W \times \frac{l}{2} \cos \theta$

Work done by this moment for an additional rotation of $d\theta = M d\theta = W \times \frac{l}{2} \cos \theta d\theta$

Total work done by the weight when it rotates through an angle ϕ

$$= \int_0^\phi W \times \frac{l}{2} \cos \theta d\theta = W \times \frac{l}{2} \sin \phi$$

When the bar becomes vertical, the angle $\phi = 90^\circ$

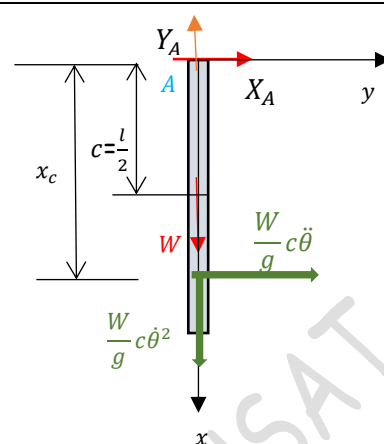
Work done by the gravity force $W = +W \frac{l}{2}$

By work energy principle, change in kinetic energy = net work done.

$$\frac{1}{2} I_A \dot{\theta}^2 = W \frac{l}{2} \gg \gg \frac{1}{2} \frac{W l^2}{g} \dot{\theta}^2 = W \frac{l}{2} \gg \gg \dot{\theta}^2 = \frac{3g}{l}$$

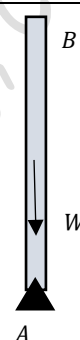
Consider the free-body diagram of the bar in the vertical position. The x-axis is drawn passing through the centre of gravity. Let us incorporate the inertia forces $\frac{W}{g}c\dot{\theta}^2$ and $\frac{W}{g}c\ddot{\theta}$ in the normal and tangential direction from the centre of percussion. Since the inertia forces are incorporated, the system is in dynamic equilibrium. So, we can write

$$\begin{aligned}\Sigma M_A = 0 & \gg \gg \frac{W}{g}c\ddot{\theta} \times x_c = 0 \gg \gg \ddot{\theta} = 0 \\ \Sigma F_x = 0 & \gg \gg X_A = 0 \\ \Sigma F_y = 0 & \gg \gg Y_A - \frac{W}{g}c\dot{\theta}^2 - W = 0 \gg \gg Y_A = \frac{5W}{2}\end{aligned}$$



Example 2

A homogeneous slender prismatic bar AB of length 1m and weight $W = 100\text{ N}$ is hinged at A as shown. If it is released from the vertical unstable equilibrium position find the reaction components at A when it swings through an angle $\theta = 30^\circ$



The bar starts from rest.

The initial kinetic energy of the bar = 0

Let $\dot{\theta}$ be the velocity of the bar when it swings through an angle $\theta = 30^\circ$

The final kinetic energy of the bar = $\frac{1}{2}I_A\dot{\theta}^2$

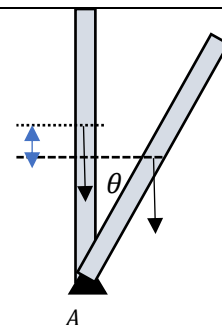
Change in kinetic energy = $\frac{1}{2}I_A\dot{\theta}^2 - 0$

Work done by the gravity force $W = +W\left(\frac{l}{2} - \frac{l}{2}\cos 30\right)$

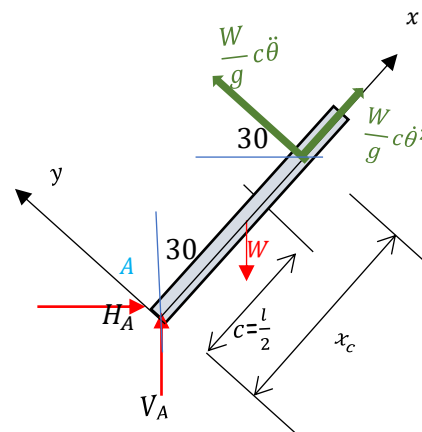
By work energy principle, change in kinetic energy = net work done.

$$\frac{1}{2}I_A\dot{\theta}^2 = W\frac{l}{2}(1 - \cos 30) \gg \gg \frac{1}{2}\frac{Wl^2}{g \cdot 3}\dot{\theta}^2 = W\frac{l}{2} \times 0.134$$

$$\dot{\theta}^2 = 3.94\text{ s}^{-2}$$



Consider the free-body diagram of the bar in the 30° inclined position. The x-axis is drawn passing through the centre of gravity. Let us incorporate the inertia forces $\frac{W}{g}c\dot{\theta}^2$ and $\frac{W}{g}c\ddot{\theta}$ in the normal and tangential direction from the centre of percussion. Since the inertia forces are incorporated, the system is in dynamic equilibrium. So, we can write the following equations.



$$I_A = \frac{Wl^2}{g3} \gg \gg \frac{Wi_A^2}{g} = \frac{Wl^2}{g3} \gg \gg \gg i_A^2 = \frac{l^2}{3}$$

$$x_c = \frac{i_A^2}{c} = \frac{\frac{l^2}{3}}{\frac{l}{2}} = \frac{2}{3}l$$

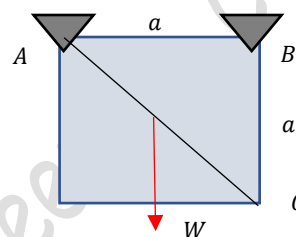
$$\Sigma M_A = 0 \gg \gg -\frac{W}{g} c\ddot{\theta} \times x_c + W \times \frac{l}{2} \sin 30 = 0 \gg \gg \ddot{\theta} = \frac{3g}{4l}$$

$$\Sigma F_h = 0 \gg \gg H_A + \frac{W}{g} c\dot{\theta}^2 \sin 30 - \frac{W}{g} c\ddot{\theta} \cos 30 = 0 \gg \gg H_A = 0.224W = 22.4 \text{ N}$$

$$\Sigma F_v = 0 \gg \gg V_A + \frac{W}{g} c\dot{\theta}^2 \cos 30 + \frac{W}{g} c\ddot{\theta} \sin 30 - W = 0 \gg \gg V_A = 0.763W = 76.3 \text{ N}$$

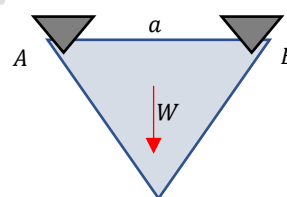
#Exercise 1

A homogeneous square plate of weight W and side a hangs in a vertical plane by two pins A and B as shown. The pin at B is removed. Calculate the reaction at A when the diagonal AC becomes vertical.



#Exercise 2

A homogeneous equilateral triangular plate of weight W and side a hangs in a vertical plane by two pins A and B as shown. The pin at B is removed. Calculate the reaction at A when centre of gravity comes directly below the axis of rotation.



Prof. Biju N, School of Engineering, CUSAT